	ZEAL EDUCATION SOCIETY'S **ZEAL INSTITUTE OF BUSINESS ADMINISTRATION, COMPUTER APPLICATION AND RESEARCH (ZIBACAR)** NARHE | PUNE -41 | INDIA **Programme – MBA & MCA**	
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Synopsis on

"MEDROUTE: EMERGENCY SMART AMBULANCE ROUTING & HOSPITAL RESOURCE DISPATCHER"

For

[NAME OF THE COMPANY]

By

PAWAR YASH VAIBHAV

AND

PATIL TUSHAR TANAJI

Submitted to

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

For the partial fulfillment of the internal credit work of

MASTERS OF COMPUTER APPLICATION (SEM – I)

Under The guidance Of

PROF. GAURAV GUPTA

Through,

Zeal Education Society's
Zeal Institute of Business Administration, Computer Application & Research (ZIBACAR)
Sr. No. 39, Narhe, Pune -411041, Phone No.:67206031
(Approved by A.I.C.T.E., New Delhi, Recognized by DTE, Govt. Maharashtra & Affiliated to S.P.P.U. Pune)

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Project Synopsis		
A.	Course Name	Masters in Computer Application (Semester I)
B.	Student's Name	Pawar Yash Vaibhav Patil Tushar Tanaji
C.	Roll No	MC2628200 MC2628192
D.	Project Title	MedRoute: Emergency Smart Ambulance Routing & Hospital Resource Dispatcher (Pune Edition)
E.	Name of Internal guide	Prof. Guarav Gupta
F.	Name of External guide	
F.	Name of Organization	
G.	Date of Submission	
Organization Profile		
1.	Name	
2.	Location	
3.	About Organization	
Project Details		
1.	Abstract	MedRoute is an operational decision-support and coordination platform that connects an ambulance, its patient telemetry, live road conditions and hospital resources in one system. Instead of sending an ambulance to the nearest hospital, MedRoute recommends the hospital that can actually treat the patient quickly, based on ICU beds, emergency beds, ventilators, blood availability and travel time. The system uses Pune's real road network (OpenStreetMap) stored in PostgreSQL with PostGIS and pgRouting. A custom in-memory graph runs the A* algorithm (with a haversine heuristic) and Dijkstra's algorithm over dynamic, time-based edge weights, so routes are recomputed when traffic changes. A weighted scoring engine ranks hospitals after hard clinical filters. Live updates reach ambulance, hospital and admin

	dashboards through WebSocket (STOMP). Patient vitals, traffic and hospital resource figures are simulated; the system does not diagnose patients.
2. Existing system	In the current practice, an ambulance driver typically heads to the nearest or most familiar hospital, often chosen by phone calls or experience. The existing approach has these limitations: 1) The nearest hospital may have no ICU bed, ventilator or blood stock, causing delay and re-transfer. 2) Hospitals receive little advance information about the incoming patient's condition and arrival time. 3) Routes are chosen by the driver and do not adapt to congestion or road blocks in real time. 4) There is no common platform linking ambulance, hospital and administration, and no record for later analysis.
3. Proposed system	MedRoute is a web-based system with three roles: ADMIN, AMBULANCE and HOSPITAL. It works as follows: 1) Routing: Pune roads are loaded into an adjacency list; A* (main) and Dijkstra (reference) compute the fastest path using cost = (length / speed) x traffic multiplier. Blocked roads are skipped and the route is recalculated from the ambulance's current node. 2) Hospital selection: hospitals failing hard filters (status, ICU bed for critical trauma, emergency bed, ventilator for low SpO2, matching blood group) are rejected; the rest are ranked by a weighted score of ETA, traffic and resource match. 3) Emergency workflow: lifecycle CREATED, AMBULANCE_ASSIGNED, PATIENT_PICKED, HOSPITAL_SELECTED, EN_ROUTE, ARRIVED,

	CLOSED, with re-evaluation when resources or traffic change. 4) Simulators: multithreaded traffic, GPS and vitals (heart rate, SpO2) simulators using thread-safe shared state. 5) Real-time dashboards: Leaflet map with route and traffic overlay, live vitals, ETA and alerts pushed to ambulance and hospital screens, plus admin reports. 6) Security: Spring Security with BCrypt and JWT-based role access.
4. Scope of the System	1) Covers emergency ambulance dispatch for the Pune area (bounding box approx. 73.75, 18.40 to 73.98, 18.65) using real OSM roads and real hospital names and locations (8-10 hospitals). 2) Includes user management, ambulance, patient, hospital and resource modules for three roles. 3) Includes dynamic routing, rerouting, hospital ranking, live telemetry, alerts and admin reports. 4) Patient vitals, traffic, ICU, ventilator and blood figures are simulated and clearly labelled as such. 5) Out of scope: medical diagnosis, live traffic APIs, mobile app and cloud deployment (listed as future enhancements).
5. Objectives of System	1) To recommend the most suitable hospital for a patient, not merely the nearest one. 2) To implement A* and Dijkstra on a real Pune road graph and compare nodes visited and time taken. 3) To model road cost as travel time with dynamic traffic multipliers so ETA is a direct output. 4) To provide live ambulance position, patient vitals, ETA and alerts to hospitals in about 1-2 seconds. 5) To give hospitals advance notice so they can prepare for the

	incoming patient. 6) To secure the system with role-based access (ADMIN, AMBULANCE, HOSPITAL) using JWT. 7) To keep emergency history and reports for analysis such as average ETA and hospital usage.
6. Environment	
a) Operating system	Windows 10/11 (development); also runs on Linux/macOS through Docker
b) Front End	React (Vite), Leaflet with OpenStreetMap tiles, SockJS/STOMP client, JavaScript, HTML, CSS
c) Back End	Java 21, Spring Boot 3.x (Spring Web, Security, Data JPA, WebSocket), JWT (JJWT), PostgreSQL 16 with PostGIS and pgRouting
Any other (please specify)	Docker (database), Maven, osm2pgrouting, OpenStreetMap data, JUnit 5 and Mockito, Postman, Git and GitHub, IntelliJ IDEA / VS Code

Date:

Sign of the student

Sign of the Internal Guide (Institute)

Sign of Project coordinator (Institute)